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A CONNECTED GLOBAL SURFACE OF SECTION BOUNDED BY ANY CHOSEN BESSE REEB ORBIT

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Editorial note. Source1008 overstates uniqueness of the dual pair for(alpha,beta)=(1,0): determinant1 requires beta-prime=1, while alpha-prime is free. The displayed(0,1) is a valid choice used for the regular-orbit model. The original wording remains.

Editorial note. Source995 writes B_i when defining the j-th solid torus Y_j; surrounding993/997 and the subsequent gluing identify the disk as B_j. The original formula remains unchanged.

Editorial note. The auxiliary local setup1070 permits negative p_0, for which its later positive-intersection/orientation assertion is not valid in general. The counted theorem1092 and its construction1125/1131 explicitly require and produce p_0>0. Only this positive-cover branch is used and endorsed in the current problem; the printed broader auxiliary wording is retained and disclosed.

Editorial note. The coordinate projection(z,s)->z in995 is not the orbit quotient for the twisted circle action1025 when alpha_j-prime is nonzero. The classical Seifert orbit quotient is allowed as background, but those two maps must not be identified. The actual positive-cover proof uses the coordinate solid-torus neighborhoods, boundary bases, explicit action and horizontal caps; the exterior Y_0 has the genuine product projection. The original pi formulas are retained: no replacement pi or new gluing proof is inserted.

Editorial note. Panel(a) of the original figure labels the large cut base region with an intersection sign. The author explicitly defines B_0-prime in1157 as B_0 minus the connecting rectangles, and that cut region is used in the subsequent sheet construction. The original figure is retained.

Allowed standard prerequisites (editorial).

- Classical Seifert surgery description of locally free circle actions on oriented three-manifolds. The classical Seifert orbit space, explicit locally free circle action and surgery/gluing conventions are allowed background, not a global-section existence theorem. Distinguish the true orbit quotient from the coordinate projection(z,s)->z printed995: the latter is not invariant under the twisted action1025. The proof uses coordinate solid tori, their boundary bases and the stated action; on the free exterior Y_0 the displayed product projection is the orbit quotient. No replacement formula for the author pi is inserted or assumed.
- Besse Reeb periodicity and negative rational Euler number. Provides the locally free action and e(Y)<0; the positive covering coefficients and global surface are still constructed.
- Equivariant local contact normal form via Moser method. The tubular form1065 is permitted; local section slope choices and all gluing remain obligations.

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- Elementary oriented-surface cut-and-paste, torus homology and Bezout/lcm arithmetic. No global-section existence theorem from Albach–Geiges is supplied as a prerequisite.

The aim of this paper is to prove some sharp inequalities involving the periods of closed orbits of Reeb flows on 3-manifolds and the contact volume. Let Y be a closed, connected, orientable 3-manifold. We recall that a one-form λ on Y is called a contact form when $\lambda \wedge d\lambda$ is nowhere vanishing. The contact form λ induces a vector field R_λ, which is called Reeb vector field of λ, by the identities $R_\lambda \lrcorner d\lambda = 0$ and $R_\lambda \lrcorner \lambda = 1$. The flow of R_λ is called the Reeb flow, and we will denote it by ϕ_λ^t. It preserves the contact form λ, and in particular the volume form $\lambda \wedge d\lambda$. Reeb flows are also called contact flows in the literature. Reeb flows on 3-manifolds constitute a special class of volume preserving flows with the remarkable feature of always having closed orbits: the Weinstein conjectures postulates that Reeb flows on arbitrary closed contact manifolds admit closed orbits, and this conjecture has been confirmed in dimension 3 by Taubes, see [34].

Borrowing once more the terminology from Riemannian geometry, a contact form λ on Y is called Besse if all its Reeb orbits are closed. Here, different Reeb orbits are not required to have the same minimal period, and therefore Besse contact forms constitute a larger class than Zoll forms. Thanks to a theorem of Wadsley [35] or, in the special case of dimension 3, an earlier theorem of Epstein [15], Besse Reeb flows are periodic (see also [33]). In our case, since Y has dimension 3, Epstein's theorem implies that all Reeb orbits of the Besse contact form λ have the same minimal period T except for finitely many ones, whose minimal period divides T. The orbits of the first kind are called regular, whereas the finitely many exceptional orbits with smaller minimal period are called singular.

0.1. **Global surfaces of section.** In this paper, a global surface of section for a contact form λ on a 3-manifold Y is a smooth map $\iota : \Sigma \to Y$, where Σ is an oriented connected compact surface with non-empty and possibly disconnected boundary, with the following properties:

- (*Boundary*) The restriction $\iota|_{\partial\Sigma}$ is an immersion positively tangent to the Reeb vector field R_λ. Namely, with the orientation on the boundary $\partial\Sigma$ induced by the one of Σ, the restriction of ι to any connected component of $\partial\Sigma$ is an orientation preserving covering map of a closed Reeb orbit of (Y, λ).
- (*Transversality*) The restriction $\iota|_{\mathrm{int}(\Sigma)}$ is an embedding into $Y \setminus \iota(\partial\Sigma)$ transverse to the Reeb vector field R_λ. In particular, the 2-form $\iota^* d\lambda$ is nowhere vanishing on $\mathrm{int}(\Sigma)$, and we assume that it is a positive area form on the oriented surface $\mathrm{int}(\Sigma)$.
- (*Globality*) For each point $z \in Y$, the Reeb orbit $t \mapsto \phi_\lambda^t(z)$ intersects Σ in both positive and negative time.

We stress that, in the literature, the notion of global surface of section may be slightly different than the one given here: for instance, the map $\iota : \Sigma \to Y$ may be required to be an embedding, or the restriction of ι to some connected component of Σ may be allowed to be an orientation reversing covering map of a closed Reeb orbit.

0.2. **The surgery description of a Besse contact form on a 3-manifold.** The proof of Theorem source: t:main will require suitable surfaces of section for the Reeb flows of contact forms sufficiently C^3-close to a Besse one. As a preliminary step, in the next subsection we shall construct global surfaces of sections for the Reeb flow of Besse contact 3-manifolds. It will be useful to employ the surgery description of

Besse contact 3-manifolds as Seifert fibered spaces, which we now recall. We refer the reader to, e.g., [30, 21] for more details.

A Seifert fibration $\pi : Y \to B$, in the generality that we need for the study of Besse contact 3-manifolds, is defined up to a suitable notion of isomorphism by a genus and $k \geq 1$ pairs of coprime integers $(\alpha_1, \beta_1), \dots, (\alpha_k, \beta_k) \in \mathbb{N} \times \mathbb{Z}$. Here, $\mathbb{N}$ denotes the set of positive integers and the coprimeness assumption implies that $\alpha_j = 1$ if $\beta_j = 0$. We denote by B_0 an oriented compact connected surface of the given genus with k boundary components. We write its oriented boundary as $\partial B_0 = \partial_1 B_0 \cup \dots \cup \partial_k B_0$, where each $\partial_j B_0$ is a connected component oriented as the boundary of B_0. Over B_0, we consider the trivial S^1-bundle

$$\pi : Y_0 := B_0 \times S^1 \to B_0, \qquad \pi(z,t) = z,$$

with its associated free S^1-action

$$t \cdot (z,s) = (z, s+t), \qquad \forall t \in S^1,\ (z,s) \in Y_0.$$

Here and elsewhere in the paper, $S^1 = \mathbb{R}/\mathbb{Z}$. Next, we consider k disjoint copies $B_j \subset \mathbb{C}$, $j = 1, \dots, k$ of the disk of some radius $\rho > 0$ centered at the origin, and the solid tori together with their base projections

$$\pi : Y_j := B_i \times S^1 \to B_j, \qquad \pi(z,t) = z.$$

By the Bézout identity, we can find pairs of coprime integers $(\alpha_j', \beta_j') \in \mathbb{Z} \times \mathbb{Z}$ such that

$$\det \begin{pmatrix} \alpha_j & \alpha_j' \\ \beta_j & \beta_j' \end{pmatrix} = 1, \qquad \forall j = 1, \dots, k. \tag{0.1}$$

The pair (α_j', β_j') is not uniquely determined by (α_j, β_j) (except if $(\alpha_j, \beta_j) = (1,0)$, in which we necessarily have $(\alpha_j', \beta_j') = (0,1)$). We introduce the oriented curves

$$\begin{aligned} m_j &= \partial B_j \times \{*\} \subset \partial Y_j, & \ell_j &= \{*\} \times S^1 \subset \partial Y_j, \\ h_j &= \{*\} \times S^1 \subset \partial_j B_0 \times S^1 \subset \partial Y_0, & f_j &= -\partial_j B_0 \times \{*\} \subset \partial Y_0. \end{aligned}$$

Here, we used the symbol $*$ to denote an arbitrary point of a space. We glue $Y_0, Y_1, \dots, Y_k$ along their boundaries by identifying

$$m_j \equiv \alpha_j f_j + \beta_j h_j, \qquad \ell_j \equiv \alpha_j' f_j + \beta_j' h_j, \qquad \forall j = 1, \dots, k$$

and denote by Y the resulting closed 3-manifold. Here, we mean that $m_j \subset \partial Y_j$ is identified with an oriented embedded circle in $\partial_j Y_0$ that is homologous to $\alpha_j [f_j] + \beta_j [h_j]$, and analogously for ℓ_j. The free S^1-action on Y_0 extends to an S^1-action on the whole Y, which on the solid tori Y_j has the form

$$t \cdot (z,s) = \big(z e^{-2\pi \alpha_j' t i}, s + \alpha_j t\big), \qquad \forall t \in S^1,\ (z,s) \in Y_j.$$

If $\alpha_j > 1$, the S^1-action is not free on the orbit

$$\gamma_j := \{0\} \times S^1 \subset Y_j,$$

and in this case we call such an orbit singular. All those S^1-orbits that are not singular are called regular. The surfaces B_j are glued accordingly to form a closed surface B. The maps π are glued as well, and the obtained $\pi : Y \to B$ is the quotient projection of the S^1-action on Y. We can always assume without loss of generality that the number of Seifert pairs (α_j, β_j) is $k \geq 2$; indeed, adding the trivial Seifert pair $(1,0)$ does not affect the Seifert fibration.

The locally free S^1-action that is induced by a Besse Reeb flow on Y can be seen as the S^1-action on the total space of a Seifert fibration $\pi : Y \to B$ as above. In this case, the Euler number

$$e(Y) := -\frac{\beta_1}{\alpha_1} - \dots - \frac{\beta_k}{\alpha_k}$$

is negative, as shown in [26] (see also [23, Th. 1.4]).

0.3. **A Global surface of section for a Besse contact form on a 3-manifold.** The existence of global surfaces of section in Seifert spaces was thoroughly investigated by Albach-Geiges [4]. In this subsection, we prove a statement which may be of independent interest (Theorem 0.1) asserting that on a Besse contact 3-manifold any given closed Reeb orbit is the (multiply covered) boundary of a surface of section as defined in Section 0.1

Let (Y, λ_0) be a Besse contact 3-manifold whose Reeb orbits have minimal common period 1, and γ_1 be an arbitrary closed Reeb orbit. The Reeb flow $\phi^t_{\lambda_0}$ defines a locally free S^1-action on Y. We can see this action as the S^1-action of a Seifert fibered structure which we describe with the same notation of the previous subsection: in particular, for each $j = 1, \dots, k$, we denote by γ_j the closed Reeb orbit corresponding to the Seifert pair (α_j, β_j). Notice in particular that we are assuming without loss of generality that our given γ_1 is the closed Reeb orbit corresponding to the Seifert pair (α_1, β_1). We recall that γ_1 has minimal period $1/\alpha_1$, and therefore it is a singular orbit if and only if $\alpha_1 > 1$.

We consider the tubular neighborhood $Y_1 \subset Y$ of γ_1, which was realized as $Y_1 \equiv B_1 \times S^1$, and under this identification we have $\gamma_1 = \{0\} \times S^1 \subset Y_1$. We equip Y_1 with the contact form

$$\lambda_0' = \frac{1}{\alpha_1}\, r^2 d\theta + \frac{1}{\alpha_1}\big(1 + 2\pi(\alpha_1'/\alpha_1) r^2\big) ds,$$

where (r, θ) are polar coordinates on the disk B_1, and $s \in S^1 = \mathbb{R}/\mathbb{Z}$. The associated Reeb vector field is given by

$$R_{\lambda_0'} = -2\pi\alpha_1' \partial_\theta + \alpha_1 \partial_s = R_{\lambda_0}|_{Y_1},$$

and therefore the associated Reeb flow agrees with the Seifert S^1-action

$$\phi^t_{\lambda_0'}(z, s) = \phi^t_{\lambda_0}(z, s) = t \cdot (z, s) = \big(z e^{-2\pi\alpha_1' t i}, s + \alpha_1 t\big).$$

Up to pulling back λ_0 by an S^1-equivariant diffeomorphism, we can assume that

$$\lambda_0|_{Y_1} = \lambda_0' = \frac{1}{\alpha_1}\, r^2 d\theta + \frac{1}{\alpha_1}\big(1 + 2\pi(\alpha_1'/\alpha_1) r^2\big) ds.$$

This can be obtained by means of a Moser's trick, see [13, Lem. 4.5]. Notice that every orbit of the Reeb flow $\phi^t_{\lambda_0}$ on Y_1 has minimal period 1, except possibly $\gamma_1 = \{0\} \times S^1$ that has minimal period $1/\alpha_1$ (the case in which γ_1 is regular is allowed, and corresponds to the Seifert invariants $(\alpha_1, \beta_1) = (1, 0)$).

For any pair of coprime integers $p_0 \neq 0$ and q_0 such that

$$\frac{q_0}{p_0} < -\frac{\alpha_1'}{\alpha_1}, \tag{0.2}$$

and for any $s_0 \in S^1$, we introduce the map

$$\iota : [0, \rho) \times S^1 \to B_1 \times S^1, \qquad \iota(r, s) = \big(r e^{2\pi(s_0 + q_0 s) i}, p_0 s\big),$$

which satisfies the following properties:

- (*Transversality*) The restriction $\iota|_{(0,\rho)\times S^1}$ is an embedding transverse to the Reeb vector field R_{λ_0}, and the image $\iota((0, \rho) \times S^1)$ intersects $-\alpha_1 q_0 - p_0 \alpha_1' > 0$ times every Reeb orbit in Y_1 other than γ_1. Therefore, the 2-form $\iota^* d\lambda_0$ is nowhere vanishing on the interior $(0, \rho) \times S^1$, and we employ it to orient the annulus $[0, \rho) \times S^1$.
- (*Boundary*) The restriction $\iota|_{\{0\}\times S^1}$ is an orientation preserving p_0-th fold covering map of the closed Reeb orbit γ_1. Here, the boundary circle $\{0\} \times S^1$ is oriented by means of the 1-form $\iota^* \lambda_0$.

We call $\iota : [0, \rho) \times S^1 \to B^2 \times S^1$ a (p_0, q_0)-local surface of section with boundary on the orbit γ_1.

We now provide the construction of a suitable global surface of section for the Besse contact 3-manifold (Y, λ) with boundary on the orbit γ_1. An alternative construction in the special case of a regular orbit in S^3 was provided by Albach-Geiges [4, Ex. 5.6]. The next result is a more precise version of Theorem source: mt:surface from the introduction.

Theorem 0.1. *Let (Y, λ_0) be a Besse contact 3-manifold with minimal common Reeb period 1, and γ_1 be any of its closed Reeb orbits. We denote by $\alpha_1 > 0$ the integer whose reciprocal $1/\alpha_1$ is the minimal period of γ_1. Then, there exist integers $b > 0$, $p_0 > 0$, and q_0 with $\gcd(p_0, q_0) = 1$, a compact connected oriented surface Σ with b boundary components and a global surface of section $\iota : \Sigma \to Y$ for (Y, λ_0) satisfying the following properties:*

(i) *Any connected component C of the boundary $\partial\Sigma$ has a collar neighborhood $A_C \cong [0, \rho) \times S^1$ such that the restriction $\iota|_{A_C}$ is a (p_0, q_0)-local surface of section with boundary on the orbit γ_1. In particular, all boundary components are positively oriented.*

(ii) *Any regular closed Reeb orbit in $Y \setminus \gamma_1$ intersects the image $\iota(\mathrm{int}(\Sigma))$ in α points, where*
$$\alpha := -\frac{b\, p_0}{e(Y)\, \alpha_1} > 0.$$
Here, $e(Y)$ is the Euler number of (Y, λ_0).

(iii) *The restriction $\iota|_{\partial\Sigma}$ is a covering map of γ_1 of degree $b\, p_0$.*

Proof. Let $(\alpha_1, \beta_1), \ldots, (\alpha_k, \beta_k)$ be the Seifert invariants of (Y, λ_0). Here, we assume without loss of generality that γ_1 is the Reeb orbit corresponding to the Seifert pair (α_1, β_1) (as we already pointed out, γ_1 is allowed to be a regular orbit, and in that case we have $(\alpha_1, \beta_1) = (1, 0)$). For every Seifert pair (α_j, β_j), we denote by (α_j', β_j') the dual pair satisfying (0.1). We set

$$\begin{aligned}
\alpha &:= \mathrm{lcm}(\alpha_2, \ldots, \alpha_k) > 0,\\
\beta &:= \left(\tfrac{\beta_2}{\alpha_2} + \cdots + \tfrac{\beta_k}{\alpha_k}\right)\alpha,\\
p &:= \beta_1\alpha + \alpha_1\beta = -e(Y)\, \alpha_1\alpha > 0, \qquad (0.3)\\
q &:= -\beta_1'\alpha - \alpha_1'\beta, \qquad (0.4)\\
b &:= \gcd(p, q),\\
p_0 &:= p/b,\\
q_0 &:= q/b.
\end{aligned}$$

By inverting the linear Equations (0.3) and (0.4), we have

$$\begin{aligned}
-\alpha &= \alpha_1 q + \alpha_1' p,\\
\beta &= \beta_1 q + \beta_1' p.
\end{aligned}$$

Notice that (0.2) is satisfied, for

$$\frac{q_0}{p_0} + \frac{\alpha_1'}{\alpha_1} = \frac{q}{p} + \frac{\alpha_1'}{\alpha_1} = \frac{\alpha_1 q + \alpha_1' p}{\alpha_1 p} = -\frac{\alpha}{\alpha_1 p} < 0.$$

Moreover,

$$\alpha = -\frac{b\, p_0}{e(Y)\, \alpha_1}.$$

With the notation of Section 0.2, we consider the small disks $B_i \subset B$ containing $\pi \circ \gamma_i$ in their interior. We connect B_1 with every other B_i by means of a rectangle R_i

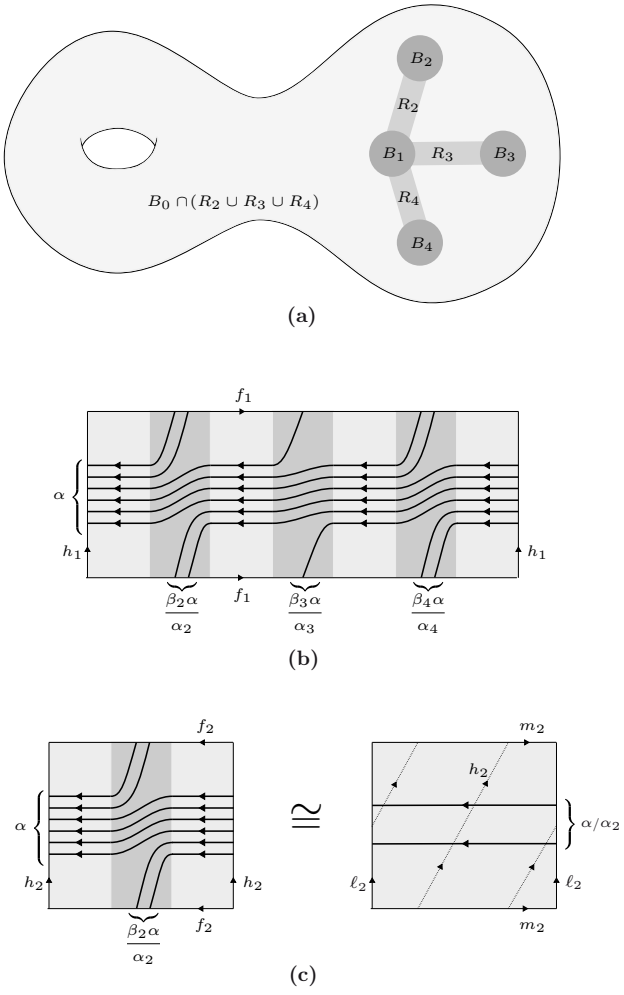


FIGURE 1. Construction of a global surface of section.

as shown in Figure 1(a). We first define the intersection of our desired surface of section with the solid torus $Y_1 = \pi^{-1}(B_1)$ as b-many (p_0, q_0)-local surfaces of section with boundary on γ_1; of course, away from their boundary on γ_1, such local surfaces of section are disjoint. We shall extend these b-many components outside Y_1 in such a way to create a (connected) global surface of section.

On the torus $\pi^{-1}(\partial_1 B_0) = \pi^{-1}(\partial B_1)$, our defined surface of section winds $-\alpha$-times around f_1 and β-times around h_1. We distribute the windings around h_1 into $k-1$ groups as in the example of Figure 1(b), where the shaded regions are the annuli $\pi^{-1}(\partial_1 B_0 \cap R_i)$, $i = 2, \dots, k$. We set $B_0' := B_0 \setminus (R_2 \cup \cdots \cup R_k)$, and extend the surface of section to $\pi^{-1}(B_0') \cong B_0' \times S^1$ as α distinct constant sections

$$B_0' \times \{t_1\}, \dots, B_0' \times \{t_\alpha\};$$

On $\pi^{-1}(R_i)$, we extend the section in $\pi^{-1}(\partial_1 B_0 \cap R_i)$ by taking the product with an interval. As a result, in each torus $\pi^{-1}(\partial B_i)$, $i = 2, \dots, k$, our surface of section winds α times around f_i, and $\alpha\beta_i/\alpha_i$ times around h_i (Figure 1(c)); this means that the surface of section winds α/α_i times around m_i, and zero times around ℓ_i.

Therefore, we can extend it to $\pi^{-1}(B_i) \cong B_i \times S^1$ as α/α_i many distinct constant sections

$$B_i \times \{t_{i,1}\}, \dots, B_i \times \{t_{i,\alpha/\alpha_i}\}.$$

It remains to show that the resulting surface of section is connected. For each $i = 2, \dots, k$, in $\pi^{-1}(\partial B_i)$ as in the left of Figure 1(c) outside the shaded region, the j-th horizontal line is connected with the $(j + \beta_i\alpha/\alpha_i)$-th one modulo α. Hence, in order the prove the claim, it suffices to remark that

$$\gcd\left(\alpha, {}^{\beta_2\alpha}/_{\alpha_2}, \dots, {}^{\beta_k\alpha}/_{\alpha_k}\right) = 1.$$

This latter equality follows from the fact that α_i and β_i are coprime and $\alpha = \operatorname{lcm}(\alpha_2, \dots, \alpha_k)$. □

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